

Sam Fine

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EDUCATION

The University of Chicago

2021 - 2025

Math B.S.; Computer Science M.S.

Chicago, IL

Accepted to concurrent BS/MS program; Grad GPA: 3.78

PUBLICATIONS & PREPRINTS

A. Kamen, **S. Fine**, B. Bhattacharyya, F. Chong, A. Goldschmidt. Comparing and correcting robustness metrics for quantum optimal control. arXiv: [2602.10349](#).

RESEARCH EXPERIENCES

Quantum Error Correction for Correlated Noise

June 2025 - Present

Advised by [Liang Jiang](#)

Chicago, IL

- Designed and simulated fault-tolerant error correction protocols for correlated noise.
- Developed machine-learning based decoders for higher-order errors in qLDPC codes.

[Robust Quantum Optimal Control](#)

May 2025 - Feb 2026

Advised by [Fred Chong](#)

Chicago, IL

- Discovered and repaired a discretization error in a standard robustness metric for quantum optimal control.
- Studied the theoretical and practical trade-offs between control complexity and error susceptibility.

Fermilab Mu2e Internship

June - Aug 2024

Advised by [Andrei Gaponenko](#)

Batavia, IL

- Developed the model of the [Mu2e experiment's](#) extinction monitor detector.
- Wrote and refactored C++ code that captured the complex geometries and constraints of the detector.

Higgs Boson Self-Coupling Classification

Jan 2023 - Jan 2024

Advised by [Philipp Windischhofer](#)

Chicago, IL

- Built and applied neural networks to extract highly-compressed, information-preserving statistics about Higgs boson self-coupling events from Large Hadron Collider datasets.

SLAC Technology Innovation Directorate Intern

2020 - 2021

Advised by [Emilio Nanni](#)

Stanford, CA

- Studied the physics of the [Cool Copper Collider](#): an advanced e^+e^- linear collider concept in the TeV class.
- Explored efficient normal-conducting particle accelerator design and built the website.

[Designed & Built a 300 KeV Cyclotron](#)

2016 - 2021

Advised by [Martin Breidenbach](#)

Stanford, CA

- Co-led project to build a 300 KeV cyclotron.
- Designed, constructed and tested vacuum, radiofrequency, ion source and detector systems.

PROGRAMMING SKILLS & INTERESTS

Languages: Python, Julia, C++

Interests: Backpacking; Biking; Reading